1.6.1 ▶ ▶ ▶ Simple Harmonic Motion

Consider a particle moving along the x axis located at a position $x(t)$. By definition, simple harmonic motion is any motion in which the position of the particle is constrained by the following differential equation:

$$\ddot{x} = -\boldsymbol{\omega}^2 x \tag{1.86}$$

where $\boldsymbol{\omega}$ is an arbitrary constant. Typically, we write this equation as depicted above, but the constant could really be written anyway one wants. With this definition, we can solve the differential equation to find how the particle moves over time. Before we do that, though, let's classify this equation. We call this kind of differential equation a **homogeneous second order linear ordinary differential equation**. Here's what all of those words mean: **Homogeneous** meaning that all terms have $x(t)$ in them or a derivative of $x(t)$. **Second Order** meaning that the highest derivative is a second derivative. **Linear** meaning that the equation is linear in $x(t)$ and its derivatives. **Ordinary Differential Equation (ODE)** meaning that there are no partial derivatives. One of the most powerful techniques to solve homogeneous linear differential equations is to guess a solution then plug it into the equation and solve for constants. In this case, we start by guessing that the solution to the equation takes the form of

$$x(t) = Ae^{\alpha t}$$

where A and α are arbitrary constants. We choose to guess this exponential solution since the derivative of e^x is also e^x, so terms often cancel. Plugging this guess into eq. 1.77 yields

$$\alpha^2 Ae^{\alpha t} = -\boldsymbol{\omega}^2 Ae^{\alpha t}$$

$$\alpha = \pm i\boldsymbol{\omega}$$

Notice that we have two values for α, meaning that we will have two solutions to the differential equation. For linear differential equations, the most general solution to a differential equation is the sum of these solutions[12]. Thus, we have

$$x(t) = Ae^{i\boldsymbol{\omega}t} + Be^{-i\boldsymbol{\omega}t} \tag{1.87}$$

with A and B both being constants. Using Euler's identity, we have

$$\begin{aligned} x(t) &= A\cos\boldsymbol{\omega}t + iA\sin\boldsymbol{\omega}t + B\cos\boldsymbol{\omega}t - iB\sin\boldsymbol{\omega}t \\ &= \boxed{C\cos\boldsymbol{\omega}t + D\sin\boldsymbol{\omega}t} \qquad (1.88) \\ &= \boxed{E\cos(\boldsymbol{\omega}t + \phi)}. \qquad (1.89) \end{aligned}$$

The two boxed solutions to the differential equation are most commonly used, since they represent sinusoidal motion. Let us analyze each of them. In the first one (eq. 1.77), setting time to $t = 0$ gives $x(0) = C$, meaning that the constant C represents the initial position of the particle. If we take a derivative of the equation, then set $t = 0$, we have $\dot{x}(0) = \boldsymbol{\omega}D$, meaning that the constant D is related to the initial velocity of the particle.

[12] I won't prove this property of linear differential equations here, so you can just take this as a fact.

Thus, if we know the initial position and velocity of the particle, we should use 1.79 to represent the motion of the particle.

Looking at eq. 1.80, we see that the maximum value of the function is E. We call this value the amplitude of the motion[13]. Setting $t = 0$ in eq. 1.80 gives $x(0) = E\cos\phi$, meaning that ϕ has some relation to the starting position of the particle. We call this value the phase of the motion. Thus, we should use this form when we want to solve for the general properties of an oscillation[14] ★

[13] As a side note, $E = \sqrt{C^2 + D^2}$, which you should try to verify using addition of sinusoids and energy

[14] Notice how we assigned physical meaning to the constants C, D, E, and ϕ, but not to A and B. Why? Because A and B could be complex numbers and have no physical meaning